

# NBA STATLINE FORECASTING

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Predicting NBA player statlines through context-aware machine learning and deployable forecasting systems.

## INTRODUCTION

### ABSTRACT

Context-aware forecasting system for NBA player statlines using 52,707 player-game observations and 70+ engineered features, evaluated under realistic pregame conditions with time-based validation.

### BACKGROUND

NBA performance varies with injuries, rotations, and matchup context. Better forecasts drive value in analytics, player props, and decision systems.

### RESEARCH QUESTIONS

- Do context-aware features outperform simple baselines and which context sources add the most predictive value?
- Which model family generalizes best and how much does feature engineering improve accuracy?

### OBJECTIVES

- Build end-to-end ETL pipeline from nba\_api data.
- Engineer 70+ context-aware pregame features.
- Compare linear and nonlinear regression models.
- Evaluate with time-based split using MAE/RMSE.
- Develop historical replay deployment prototype.

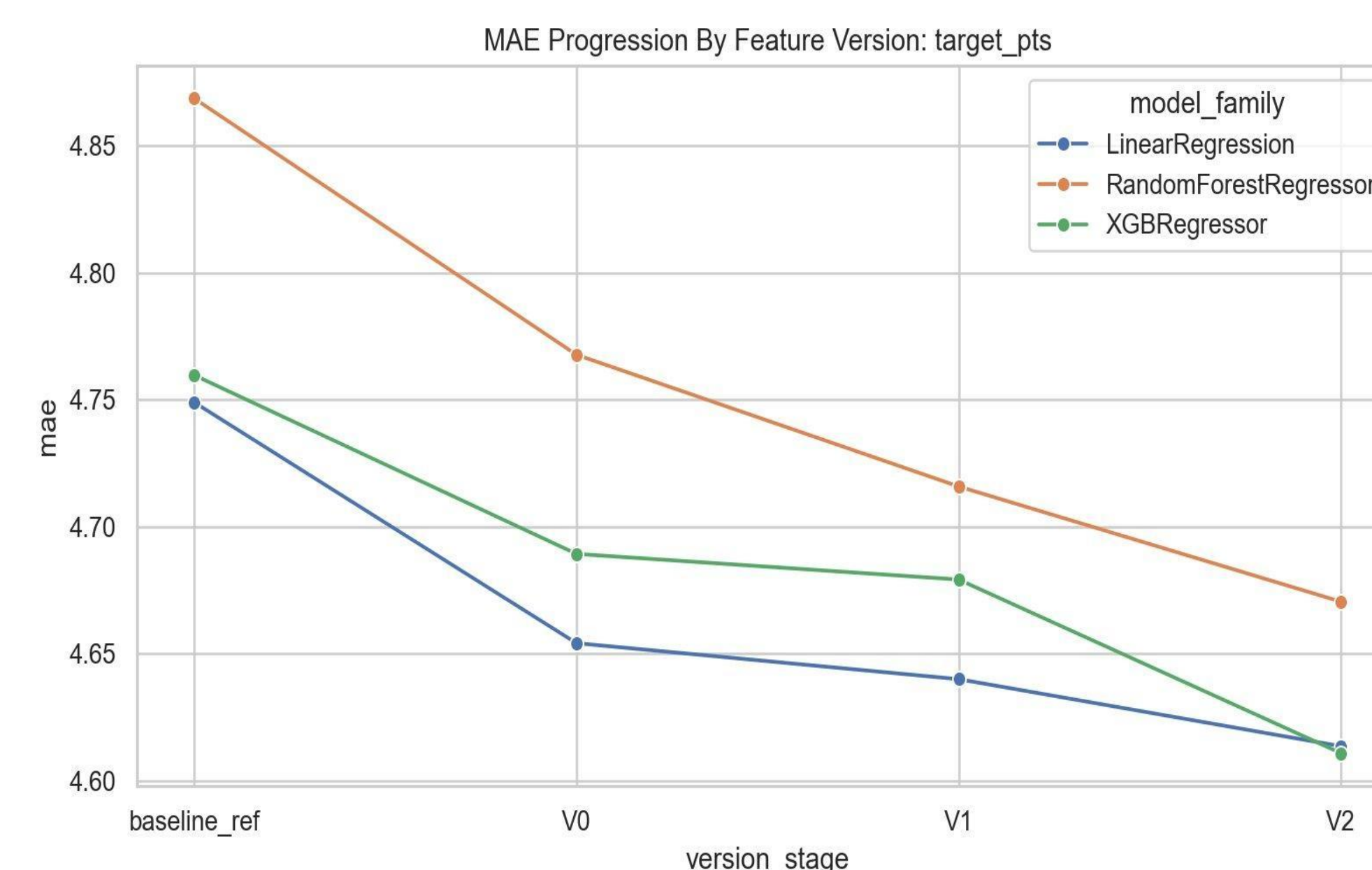
## DATA PIPELINE

### DATA COLLECTION

Data sourced via nba\_api: player game logs, team stats, and opponent context across 2023-25 seasons. 52,707 rows, 694 players, 2,460 games. Train: 2023-24 (26,401). Eval: 2024-25 (26,306).

### FEATURE ENGINEERING

Season averages capture baseline ability. L5/L10 metrics reflect form. Teammate absences and opponent pace/defense add context. All features are pregame only.



All models improved from baseline to V2. LR achieved strongest final performance.

### MODELS

- Linear Regression
  - Random Forest
  - XGBoost
- Compared interpretable linear methods against nonlinear ensemble methods.

### EVALUATION METHODOLOGY



Time-based holdout: train 2023-24, test on unseen 2024-25.



MAE and RMSE measure forecast accuracy and large-error risk.

### RESULTS

LR and XGBoost V2 achieved best PTS MAE (4.61), outperforming baseline and RF. Context versions (V0-V2) consistently reduced error. Simpler models matched ensembles, indicating strong features captured most predictive signal.

|                    | MAE  | RMSE |
|--------------------|------|------|
| Linear Reg (V2)    | 4.61 | 5.98 |
| XGBoost (V2)       | 4.61 | 6.00 |
| Random Forest (V2) | 4.67 | 6.06 |
| Baseline           | 4.75 | 6.20 |

### VERSIONS

- Baseline A: season averages
- Baseline B: rolling (L5/L10) averages
- V0: baseline + opponent + home/away
- V1: V0 + availability/missing production
- V2: V1 + usage + matchup context

## CONCLUSIONS

### KEY FINDINGS

Well-engineered features improved all model families. LR achieved lowest MAE (4.61), suggesting strong features reduced the need for model complexity. Gains were consistent across PTS, REB, and AST.

### FUTURE WORK

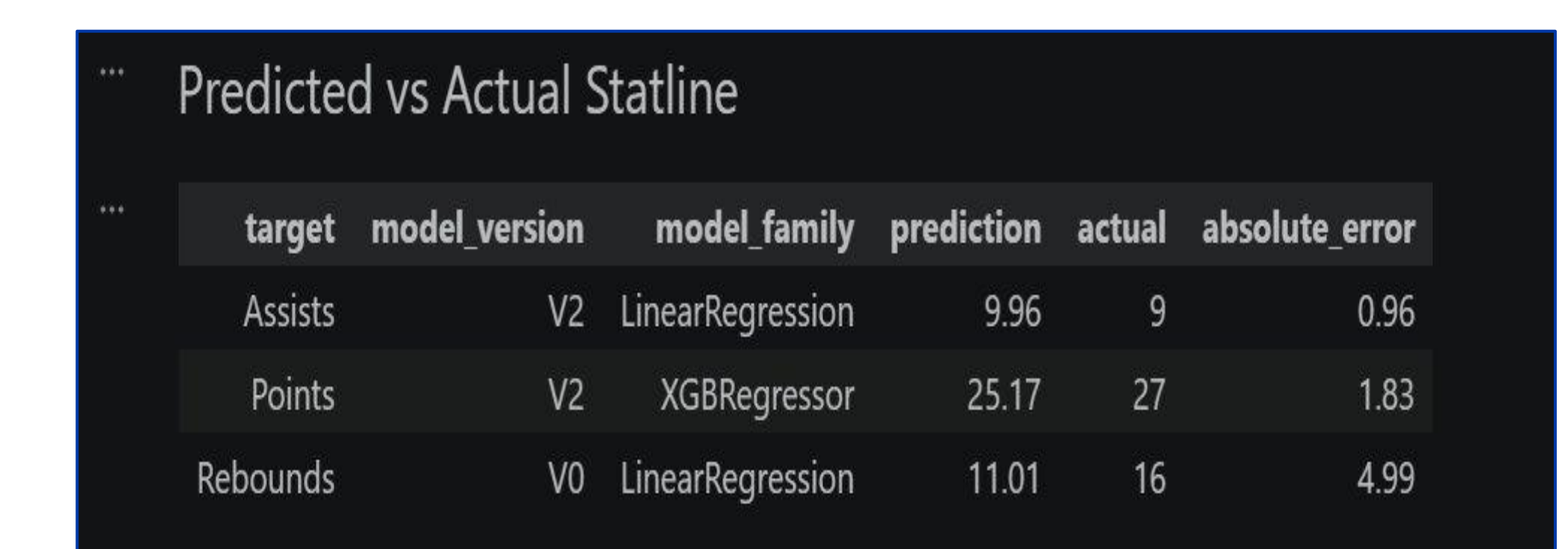
- Hyperparameter optimization (RF/XG).
- API/dashboard for live forecasts.
- Real-time injury/lineup feeds.
- Rolling retraining during season.
- PRA / 3PM market expansion.

### MONITORING & GOVERNANCE

Track rolling MAE/RMSE vs benchmark. Trigger review when MAE rises 5%+ over baseline. Monitor feature drift and lineup shifts. Experiments versioned via Weights and Biases and GitHub.

### DEPLOYMENT

Replay prototype forecasts held-out games using only pre-tipoff data, simulating live and realistic market deployment.



| target   | model_version | model_family     | prediction | actual | absolute_error |
|----------|---------------|------------------|------------|--------|----------------|
| Assists  | V2            | LinearRegression | 9.96       | 9      | 0.96           |
| Points   | V2            | XGBRegressor     | 25.17      | 27     | 1.83           |
| Rebounds | V0            | LinearRegression | 11.01      | 16     | 4.99           |

Jokic (11/2/24): Pregame forecast vs actual statline using held-out 2024-25 data.